

Public Economics

Lecture 4 · Taxation

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Today's Questions

- What are the different types of taxes governments levy, and how does the burden of a tax actually get shared between buyers and sellers?
- What does it mean for a tax system to be “fair,” and how do we measure that?
- Why does every real-world income tax deviate from a clean, comprehensive definition of income — and are those deviations justified?

Based on

Gruber, J. *Public Finance and Public Policy*, Ch. 18 (§18.1, 18.3–18.5); Leenders, W., *Microeconomics of Public Policy, Part IV: Taxation* (Sciences Po, 2023–2024).

4.1 Types of Taxation

4.2 Tax Incidence

4.3 Measuring the Fairness of Tax Systems

4.4 Defining the Income Tax Base

The Rise of the Fiscal State

The Distribution of the Tax Burden

Conclusion

4.1 Types of Taxation

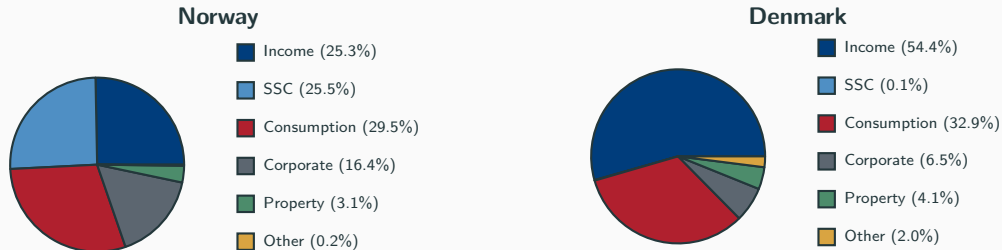
Five Types of Taxes

- **Payroll tax:** levied on earnings from work; finances social insurance (pensions, unemployment, health).
- **Individual income tax:** paid on income accrued during the year — a broader base than payroll (interest, etc.) and often assessed per household, not per worker.
- **Corporate income tax:** levied on corporate earnings, to catch capital income that might otherwise escape individual-level taxation.
- **Wealth taxes:** levied on the *value of assets* held (not income), e.g. property tax, estate/inheritance tax.
- **Consumption taxes:** levied on spending, e.g. VAT/sales tax; when applied to one specific good (fuel, tobacco, alcohol), called an **excise tax**.

Payroll, income, and wealth taxes are **direct** taxes (assessed on individuals); consumption taxes are **indirect** (assessed on transactions).

Source: Gruber, *Public Finance and Public Policy*, §18.1.

International Tax Revenues by Type



2018 — two very different OECD tax mixes: Norway spreads revenue across income, payroll, consumption, and corporate taxes evenly; Denmark leans heavily on individual income tax.

Source: Gruber, *Public Finance and Public Policy*, Fig. 18-2, citing OECD (2020).

Where Does Austria Fit?

- Recall from Lecture 1: Austria's tax mix leans heavily on **labor** — VAT ($\approx$ €35.4bn, 2022), wage tax ($\approx$ €25bn), and social contributions dominate; corporate tax is comparatively small ($\approx$ €7.8bn, only 2.2% of GDP).
- Austria is closer to the **OECD average** pattern above than to Denmark's income-tax-heavy mix — consumption and payroll taxes both play a substantial role.

A structural fact worth remembering

Austria abolished its general **wealth tax** (*Vermögenssteuer*) in 1994, and its **inheritance/gift tax** lapsed in 2008 after the *Verfassungsgerichtshof* ruled the property-valuation method unconstitutional and no replacement was passed. Austria has had *no* wealth or inheritance tax since.

Source: Lecture 1; BMF; *Verfassungsgerichtshof* case law on the Erbschafts- und Schenkungssteuergesetz (2007/2008).

4.2 Tax Incidence

Tax Incidence: Who Really Pays?

Tax incidence

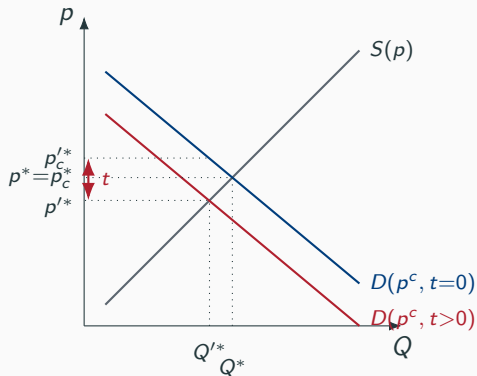
The study of how taxes affect the prices buyers and sellers face — and therefore who really bears the burden.

- **Legal (statutory) incidence:** who is legally required to send the tax payment to the government.
- **Economic incidence:** who actually ends up worse off, once prices adjust.

These are *not* the same thing — and which one matters is one of the most counterintuitive results in public finance.

Source: Leenders, *Microeconomics of Public Policy*, Part IV (Sciences Po).

The Canonical Case: An Excise Tax

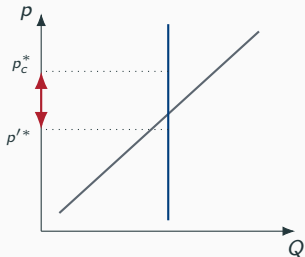


A tax t on consumers wedges $p_c = p + t$ apart from p . Quantity falls from Q^* to Q'^* ; the consumer price rises by *less than* t — both sides share the burden.

Source: Following Leenders, *Microeconomics of Public Policy*, Part IV (Sciences Po).

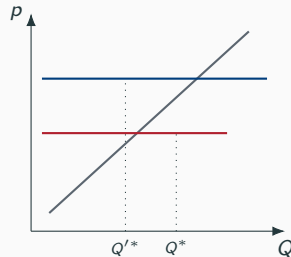
Two Extreme Cases

Perfectly inelastic demand



Quantity can't change — consumers bear **100%** of the burden.

Perfectly elastic demand



Consumer price can't change — producers bear **100%** of the burden.

The side of the market with the *less elastic* response bears more of the tax — regardless of who writes the check.

Elasticity Determines Incidence

- In theory, incidence does *not* depend on who is statutorily required to pay — a tax on sellers and an equivalent tax on buyers have the same economic incidence.
- What matters is the **relative elasticity of supply and demand**:

$$\varepsilon_D = \frac{p_c}{D} \frac{dD}{dp_c}, \quad \varepsilon_S = \frac{p}{S} \frac{dS}{dp}$$

Application: excise taxes in Austria

Austria taxes tobacco, alcohol, mineral oil, and (since 2022) CO₂ emissions via excise duties. Demand for cigarettes and heating fuel is famously **inelastic** in the short run — so standard incidence theory predicts these taxes fall mostly on *consumers*, not producers.

The Equity–Efficiency Trade-off, Revisited

A progressive income tax has two effects:

1. **Equity gain:** reduces income inequality.
2. **Efficiency loss:** people may work less when income is taxed — if they care about *net*, not gross, pay.

The size of the efficiency loss is governed by the **elasticity of labor supply** (Lecture 2).

Governments can dampen it by using tax revenue to subsidize goods complementary to working, such as child care. $\Rightarrow$ “How Can Scandinavians Tax So Much?” (Kleven, 2014) — high-tax Nordic countries also invest heavily in making work easy (child care, low-distortion consumption taxes), which is part of how they sustain such high tax revenue with limited efficiency loss.

Source: Kleven, H., *Journal of Economic Perspectives* 28(4), 2014, cited in Leenders, Part IV.

4.3 Measuring the Fairness of Tax Systems

Average vs. Marginal Tax Rates

Marginal tax rate

The share of the *next euro* of taxable income paid in tax.

Average tax rate

Total tax paid, divided by total (gross) income — a weighted average of all the marginal rates paid along the way.

Austria's 2024 Income Tax Brackets

Taxable income (€)	Marginal rate
0 – 12,816	0%
12,816 – 20,818	20%
20,818 – 34,513	30%
34,513 – 66,612	40%
66,612 – 99,266	48%
99,266 – 1,000,000	50%
above 1,000,000	55%

The 55% top bracket is a temporary surcharge

(legislated 2016–2029). Brackets are progressive: each additional euro is taxed at a higher rate than the last.

Source: Bundesministerium für Finanzen (BMF), Einkommensteuertarif 2024.

Worked Example: Lena's Tax Bill

Lena has taxable income of €80,000. Walking up the bracket schedule:

$$(\text{€}8,002 \times 0.20) + (\text{€}13,695 \times 0.30) + (\text{€}32,099 \times 0.40) + (\text{€}13,388 \times 0.48) = \text{€}24,974.74$$

- Lena's **marginal** tax rate is **48%** — the rate on her next euro earned.
- Lena's **average** tax rate is $\text{€}24,974.74 / \text{€}80,000 = \mathbf{31.2\%}$.

The gap between the two — 48% vs. 31.2% — is entirely due to progressivity: the first €12,816 was tax-free, and every bracket below her marginal one was taxed at a lower rate.

Vertical and Horizontal Equity

Vertical equity

Groups with more resources should pay higher taxes than groups with fewer resources.

Horizontal equity

Individuals who are similar, but make different choices, should be treated the same by the tax system.

- A **progressive** system: average tax rates *rise* with income.
- A **proportional** (flat) system: average tax rates are *constant*.
- A **regressive** system: average tax rates *fall* with income.

When “Fair” Causes a Riot

In March 1990, rioters in London set fire to parked cars and smashed storefronts, injuring more than 400 people. The cause: Margaret Thatcher's proposal to replace property taxes with a **poll tax** — a flat charge levied *equally* on every adult, rich or poor.

Why it provoked outrage

The tax burden shifted away from wealthy property owners and onto poorer citizens who hadn't previously paid property tax at all — a textbook violation of **vertical equity**. Thatcher was ousted as Conservative leader soon after; the tax was abandoned.

(England's last attempt at such a tax, in 1381, ended with the beheading of several prominent officials.)

Source: Gruber, *Public Finance and Public Policy*, §18.3.

The problem

With fixed, progressive brackets, **inflation** pushes people into higher brackets even when their *real* income hasn't grown — a stealth tax increase known as *kalte Progression* (bracket creep).

- Effective 1 January 2023, Austria began **automatically indexing** tax brackets to inflation: two-thirds of the inflation adjustment happens automatically by law each year; the remaining third is allocated at the government's discretion via a separate “*Progressionsabgeltungsgesetz*.”
- The top €1,000,000 / 55% bracket is *excluded* from indexation.

Discussion: is partial (two-thirds automatic) indexation a reasonable compromise, or does the discretionary third just recreate the same political problem it was meant to fix?

Source: BMF; WKO; Parlament.gv.at, Teuerungs-Entlastungspaket II (2022).

4.4 Defining the Income Tax Base

The Haig-Simons Comprehensive Income Definition

Haig-Simons income

Taxable resources = an individual's *ability to pay* = total consumption during the year
+ any increase in wealth.

- In principle, comprehensive income improves *both* vertical equity (all resources are taxed, however received) *and* horizontal equity (the form compensation takes shouldn't matter).
- In practice, every real tax system deviates from this ideal — the question is whether each deviation is justified.

Source: Gruber, *Public Finance and Public Policy*, §18.4.

Austria vs. the US: In-Kind Compensation

- The classic Haig-Simons violation in the US textbook: employer-provided **health insurance** is not taxed as income, even though it clearly raises a worker's ability to consume.
- Austria takes a stricter approach for many in-kind benefits: the *Sachbezugswerteverordnung* taxes the value of a company car, subsidized housing, and other in-kind perks as part of wage income.

Not every deviation from Haig-Simons is universal across countries — the “right” tax base is a policy choice, not a law of nature.

Deviations for Ability-to-Pay and Earning Costs

- **Ability-to-pay deviations:** Austria's *außergewöhnliche Belastungen* (extraordinary burden deductions) let taxpayers deduct large, involuntary costs — medical expenses, disaster losses — on the same logic as Gruber's “fire damage” example: these expenditures aren't really *consumption*, so they shouldn't count as ability to pay.
- **Cost-of-earning-income deviations:** Austria's *Werbungskosten* let workers deduct job-related expenses, including the *Pendlerpauschale* (commuter allowance, Lecture 1) — the same logic as deducting business costs from Haig-Simons income.

Both are easy to justify in principle — and hard to draw a clean line around in practice (was that lunch a business cost, or consumption?).

Housing: A Case Where Austria Diverges Sharply

- Whether you rent or own, you consume €X/month of housing services — your *imputed rental value*. A true Haig-Simons tax would treat renters and owners identically.
- The **US** does not tax imputed rental income, *but* lets owners deduct mortgage interest — a large, one-sided subsidy to owning ($\approx$ €80bn/year in forgone US federal revenue) that the textbook's own cited research (Glaeser & Shapiro, 2003; a later study finding a Danish policy change had a *precisely-estimated zero effect* on homeownership) struggles to justify on positive-externality grounds.

Austria took a different path

Austria has **no general mortgage-interest deduction**. A narrower *Sonderausgaben* deduction for housing construction/renovation loans was closed to new contracts in 2016 and phased out entirely by the 2021 tax-assessment year — Austria now taxes owners and renters far more symmetrically than the US does.

Source: Gruber, *Public Finance and Public Policy*, §18.5; findok.bmf.gv.at; Statistik Austria (2021 Arbeitnehmerveranlagung).

Deductions vs. Credits

Tax deduction

Reduces *taxable income* — its value to the taxpayer is $\$(1 - t)$ per euro, so it's worth *more* to people in higher brackets.

Tax credit

Reduces *tax owed* directly — worth the same to everyone, regardless of bracket, and thus more progressive.

A **refundable** credit goes further still: it pays out even to people who owe little or no tax — turning the tax system into a delivery mechanism for cash transfers to low earners.

Austria's Own Refundability Debate: *Familienbonus Plus*

Introduced in 2019 as a tax **credit** (not a deduction) for families with children.

- 2024 value: €2,000.16/year per child under 18; €700.08/year per child 18 or older.
- Because it's a credit against tax owed, it delivers *nothing* to a parent who owes little or no income tax — exactly the refundability problem the US Child Tax Credit debate raises.
- Austria's partial fix: the *Kindermehrbetrag* (2024: €700/child) tops up low earners who claim the sole-earner/single-parent credit, closing the gap between their (near-zero) tax liability and €700.

Discussion: does the *Kindermehrbetrag* fully solve the vertical-equity problem of a non-refundable credit, or only partly?

Source: Bundeskanzleramt; BMF, Familienbonus Plus & Kindermehrbetrag (2024).

Tax expenditures

Revenue the government forgoes through special exclusions, deductions, credits, or preferential rates — functionally equivalent to direct spending, just routed through the tax code instead of the budget.

In the US, tax expenditures total over €1 trillion/year — more than 40% of total tax revenue. Every deduction and credit on the last several slides (Familienbonus Plus, Werbungskosten, außergewöhnliche Belastungen) is, by this definition, an Austrian tax expenditure too.

Source: Gruber, *Public Finance and Public Policy*, §18.5, citing U.S. Department of the Treasury (2021a).

The Rise of the Fiscal State

How Did We Get Progressive Taxation?

- Top marginal income-tax rates were in the single digits almost everywhere before 1900 — then rose sharply, above all in the US, UK, Sweden, Germany, and France, from the 1910s onward.
- **World War I** acted as the catalyst: many countries raised top rates on income and inheritances explicitly as a “conscription of capital” to match the conscription of soldiers.
- The wartime rate hikes only *stuck* because of lasting technological and institutional changes: employer-based withholding, and — eventually — the end of bank secrecy.

Austria's own version of that last point

Austria was long known for strict *Bankgeheimnis* (bank secrecy). That changed with the OECD's Common Reporting Standard: Austria began automatically exchanging account data with foreign tax authorities starting September 2017.

Source: Leenders, Part IV, citing Piketty, *Capital and Ideology* (2020); Gemeinsamer Meldestandard-Gesetz (GMSG), effective 2016–2017.

Why This History Matters

- The rise in top rates on the wealthy over the twentieth century is a major part of the story behind the dramatic *fall* in income and wealth inequality across the same period (Lecture 1).
- Total tax revenue as a share of national income rose alongside it, from roughly 5–10% (1870) to 30–55% (today) across today's rich democracies — with the US consistently at the *low* end of that range and the Nordic countries at the *high* end.

This isn't ancient history — it's the direct backdrop to the debates in this lecture: what should be taxed, how progressively, and how the tax base should be defined.

Source: Leenders, Part IV, citing Piketty (2020) and long-run OECD/WID revenue series.

The Distribution of the Tax Burden

Who Actually Pays the Taxes?

- Rather than asking how a *hypothetical* tax change would affect prices (incidence theory), a newer research tradition (Saez & Zucman, 2019, and successors) asks: given the tax system *as it exists*, what share of everyone's income actually goes to tax, by income rank?
- Applied to the Netherlands, Italy, France, and the US, this research finds effective tax rates that **rise** through the middle of the distribution — then **flatten or fall** at the very top.

The progressivity we build into income-tax *brackets* doesn't automatically show up in the effective rate actually paid, once every tax is counted.

Source: Leenders, Part IV, citing Saez & Zucman, *The Triumph of Injustice* (2019) and related Blanchet-Saez-Zucman-style distributional tax studies.

Four Insights on Who Pays

1. **Income taxes** are progressive for the bottom 90% of earners, but play only a minor role at the very top — where income increasingly comes from capital, not wages.
2. **Corporate taxes** act as a kind of minimum tax on the super-rich, since a large share of top-end income flows through corporate ownership.
3. **Consumption taxes** hit lower-income households hardest, because they consume a larger *share* of their income than the wealthy do.
4. **Payroll taxes** are levied only on labor income — and labor income's share of total income *falls* as you move up the distribution, which is regressive.

Austria's own version of insight #4

Austria's social-insurance contributions are capped at the *Höchstbeitragsgrundlage* — €6,060/month in 2024. Above that, an extra euro of salary pays *zero* additional social-insurance contribution — a regressive payroll tax at the top of the distribution.

Source: Leenders, Part IV; sozialversicherung.at, Höchstbeitragsgrundlage 2024.

Conclusion

Highlights

- Taxes come in five broad types (payroll, individual income, corporate income, wealth, consumption) and countries mix them very differently — Austria leans on labor income, much like the OECD average.
- Economic incidence — who really bears a tax — depends on relative elasticities, not on who legally writes the check.
- Fairness has two distinct dimensions (vertical, horizontal equity), and every real tax system navigates trade-offs between them, often visibly and politically (Thatcher's poll tax; Austria's *kalte Progression* reform).
- No real income tax base matches the Haig-Simons ideal — Austria and the US deviate from it in different, and sometimes opposite, ways (in-kind pay, housing).
- Whether progressive brackets translate into a progressive *effective* tax burden depends on the whole system — including payroll-tax caps like Austria's own *Höchstbeitragsgrundlage*.

Discussion Questions for Next Time

1. Austria has no wealth or inheritance tax, while several EU neighbors do. What would the efficiency and equity trade-offs be in reintroducing one?
2. If demand for heating fuel is highly inelastic, who really bears Austria's CO₂ tax — and does that change your view of whether it's a fair policy?
3. Is Austria's decision to tax company cars and other in-kind benefits (Sachbezug), while the US doesn't tax employer health insurance, better tax policy — or just a different set of political trade-offs?
4. Austria's Familienbonus Plus is a non-refundable credit, topped up by the Kindermehrbetrag. Design an alternative policy that achieves the same goal (support for families with children) while being more/less redistributive — and justify your choice.
5. If effective tax rates flatten at the very top of the income distribution (as Saez-Zucman-style research finds), what does that imply about relying on the income tax alone to address wealth concentration?

- Textbook: Gruber, J. *Public Finance and Public Policy*, Ch. 18 (§18.1, 18.3–18.5), Worth Publishers.
- Leenders, W., *Microeconomics of Public Policy, Part IV: Taxation*, lecture slides, Sciences Po (MPA 2023–2024) — tax incidence theory, the rise of the fiscal state, and the distribution of the tax burden, citing Kleven (2014), Piketty (2020), Saez & Zucman (2019), and Smith, Yagan, Zidar & Zwick (2019).
- Bundesministerium für Finanzen (BMF): Einkommensteuertarif 2024; Familienbonus Plus & Kindermehrbetrag (2024); Gemeinsamer Meldestandard-Gesetz (GMSG).
- sozialversicherung.at: Höchstbeitragsgrundlage 2024.
- findok.bmf.gv.at; Statistik Austria (2021 Arbeitnehmerveranlagung) — housing-loan Sonderausgaben phase-out.
- *Verfassungsgerichtshof*, case law on the Erbschafts- und Schenkungssteuergesetz (2007/2008); WKO; Parlament.gv.at, Teuerungs-Entlastungspaket II (2022).

Thank you!

Questions and discussion